供應鏈營運分析 Supply Chain Operations

整合訂單需求、工廠產能、運費覆蓋率與營運成本

Executive Overview

快速掌握訂單需求、運費覆蓋率與整體成本

View Overview →

Plant & Capacity

比較各工廠訂單量、每日產能與產能利用率

View Capacity →

Freight & Cost

分析費率配對、未配對訂單與預估運輸成本

View Freight →

Customer & Product

比較 VMI 顧客需求，辨識單一工廠供應風險

View Details →

分析目的
評估產能負荷、需求集中度與費率覆蓋情況

資料範圍
本報表分析 9,215 筆單日訂單，涵蓋工廠、顧客、產品與運送資料

商業價值
辨識產能瓶頸、費率缺口與成本改善機會

分析工具
Python 、MySQL、 Power BI

資料來源：Supply Chain Dataset

作者: 鄒雨君

Supply Chain Executive Overview 供應鏈營運總覽

訂單需求 Order Demand, 運費覆蓋率 Freight Coverage, 營運成本摘要 Operational Cost Summary

Order Date: May 26, 2013

Coverage excludes 854 CRF customer-arranged orders

Total Orders	Total Units	Active Plants	Unique Customer	Unique Products	Freight Coverage Rate
9,215	29.51M	7	46	772	83.61%

Home

Executive Overview

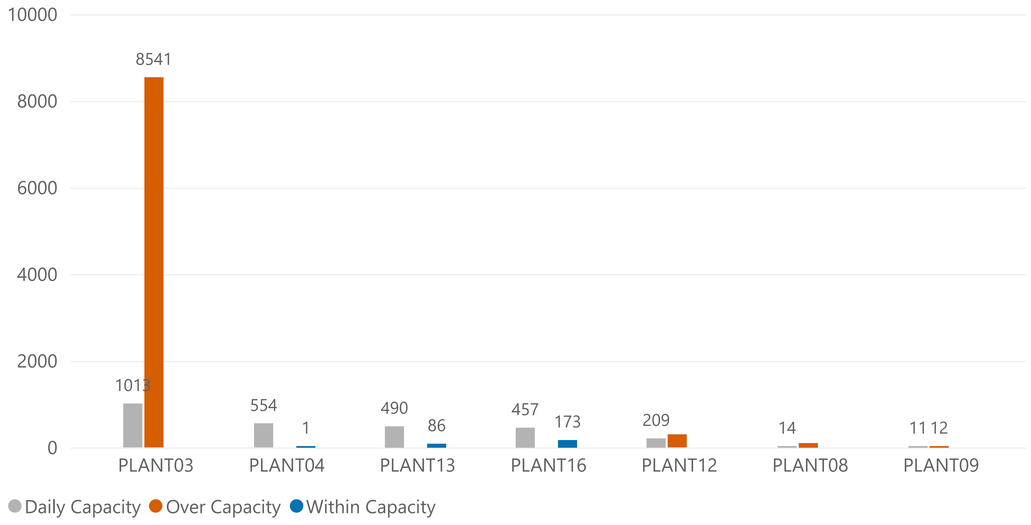
Plant & Capacity

Freight & Cost

Customer & Product

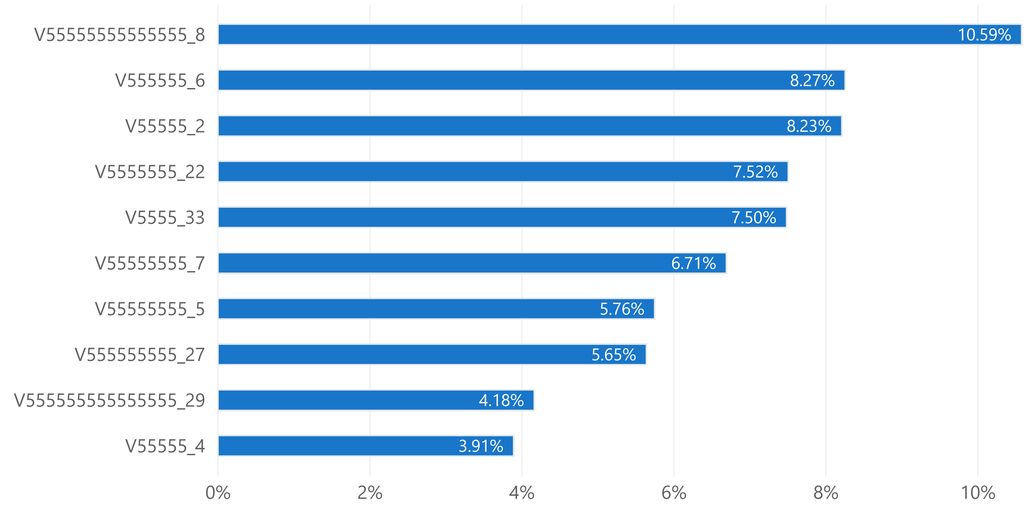
Order Volume vs. Daily Capacity by Plant

哪些工廠的訂單量超過每日產能?



Top Customers by Order Share

哪些顧客帶來最多訂單?



Supply Network Concentration

Category	Primary Component	Total Orders	Order Share
Carrier	V444_0	6264	67.98%
Plant	PLANT03	8541	92.69%
Route	PORT04 → PORT09	9041	98.11%

Key Insights

- PLANT03 承擔 92.69% 的訂單，工廠需求高度集中。
- PORT04 → PORT09 處理 98.11% 的訂單，存在主要路線依賴風險。
- V444_0 承擔 67.98% 的訂單，承運商使用亦呈現集中。
- Top 10 顧客貢獻 68.31% 的訂單，需求主要來自一群高量顧客。

Plant & Capacity 工廠與產能

工廠訂單需求、每日產能與倉儲成本分析

Select Plant

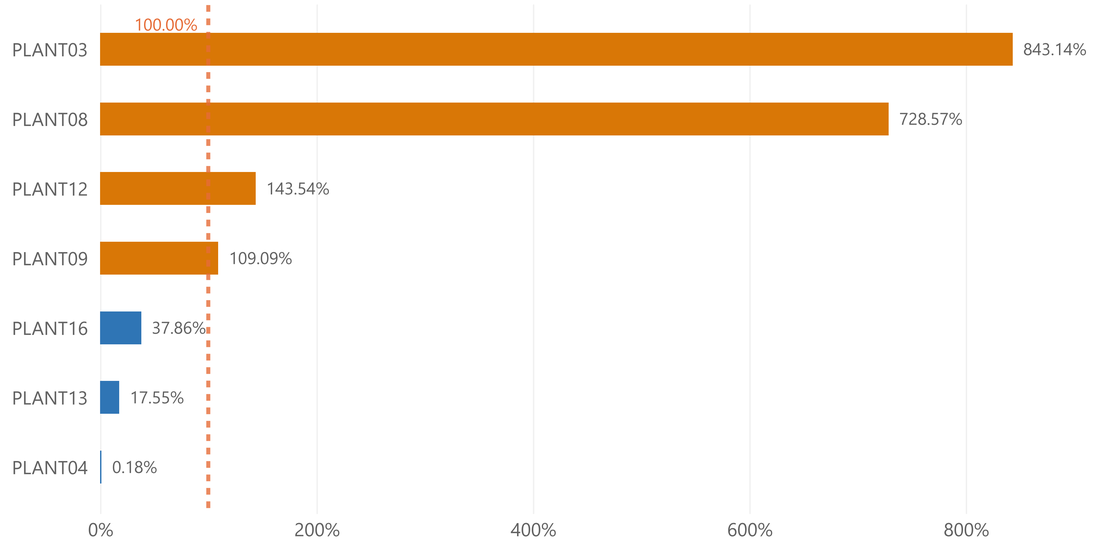
- ☐ (Blank)
- ■ PLANT03
- ☐ PLANT04
- ☐ PLANT08
- ☐ PLANT09
- ☐ PLANT12
- ☐ PLANT13
- ☐ PLANT16

產能狀態	當日訂單	預估倉儲成本
Capacity Status 超過產能 | Over Capacity	Selected Plant Orders 8,541	Estimated Warehouse Cost $14.81M

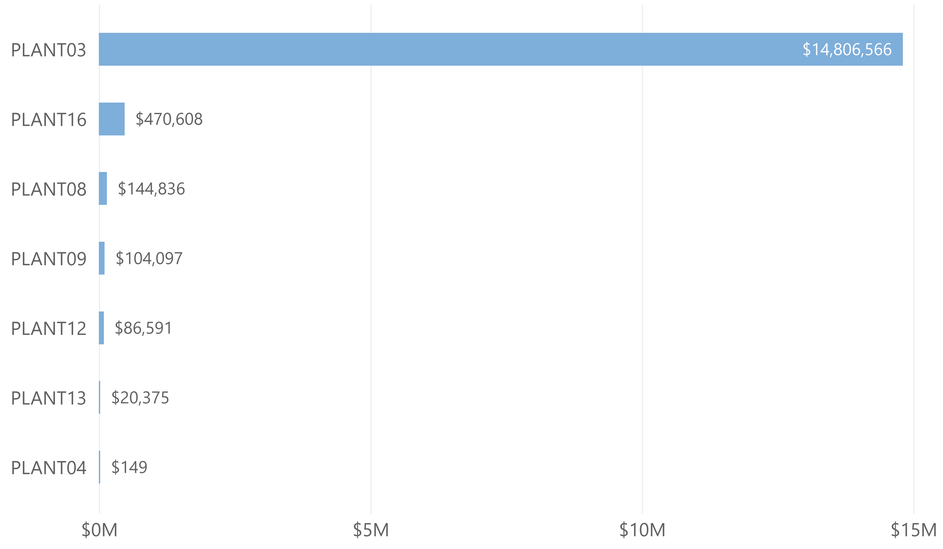
產能利用率	每日訂單產能	當日商品數量
Capacity Utilization % 843.14%	Plant Daily Order Capacity 1,013	Selected Plant Units 29M

- 產能利用率 = 當日訂單數 ÷ 每日訂單產能
- 預估倉儲成本 = 商品數量 × 單位倉儲成本
- 資料僅涵蓋 2013/05/26；產能結果為情境比較，不代表長期營運表現

Capacity Utilization by Plant



Estimated Warehouse Cost by Plant



Home
Executive Overview
Plant & Capacity
Freight & Cost
Customer & Product

Freight & Cost 運費與成本分析

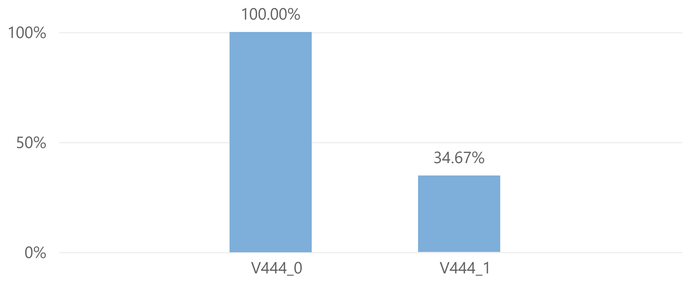
分析運費費率覆蓋、未配對訂單與承運商成本

費率覆蓋率	成功配對	未配對訂單	總運費
Freight Coverage Rate	Matched Orders	Unmatched Orders	Total Freight Cost
83.61%	6,991	1,370	$69,568.23

覆蓋率以 8,361 筆適用訂單計算，已排除 854 筆由顧客安排運輸的 CRF (顧客自行安排運輸)訂單。

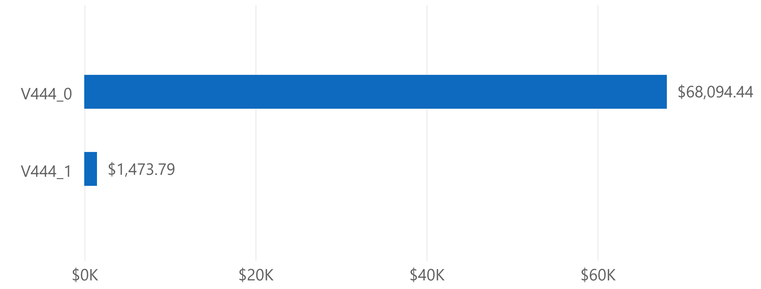
Freight Coverage by Carrier

哪個承運商的費率覆蓋不足?



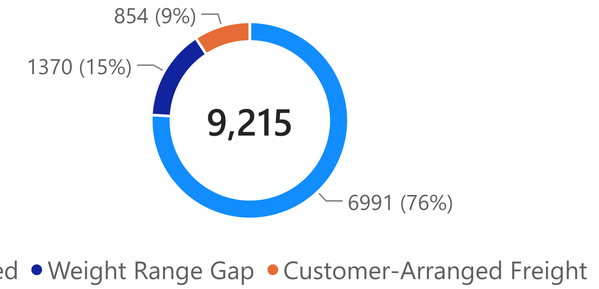
Freight Cost by Carrier

已配對運費成本集中在哪個承運商?



Freight Matching Status

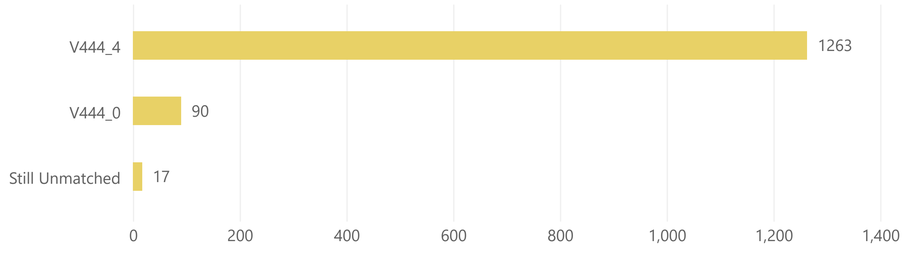
訂單的運費配對結果如何分布?



Improvement Scenario 覆蓋率優化方案

Recovered Orders by Alternative Carrier

1,370 筆費率缺口可如何重新分配?



優化後覆蓋率	最終覆蓋訂單	新增運費成本
Optimized Coverage Rate	Final Covered Orders	Additional Freight Cost
99.80%	8,344	$32,185.5

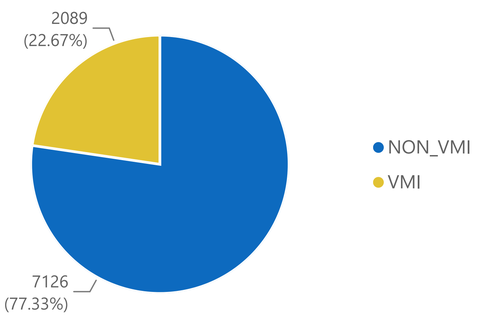
目前費率缺口主要集中於 V444_1。將未配對訂單改配最低成本的替代承運商後，可補足 1,353 筆訂單，將覆蓋率由 83.61% 提升至 99.80%，但預估增加運費 $32,185.50；導入前應確認主要替代承運商 V444_4 的承運能力與服務條件。

Customer & Product 顧客與商品分析

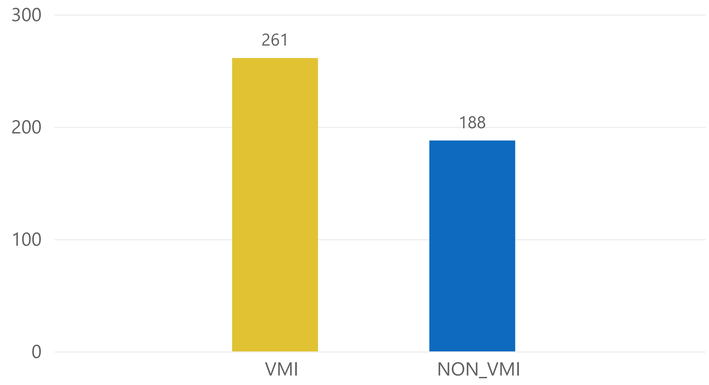
顧客需求、商品供應風險

VMI Customer Share	VMI Order Share	Count of product_id	VMI Customers
17.39%	22.67%	772	8

Orders by Customer Type
各類顧客訂單量



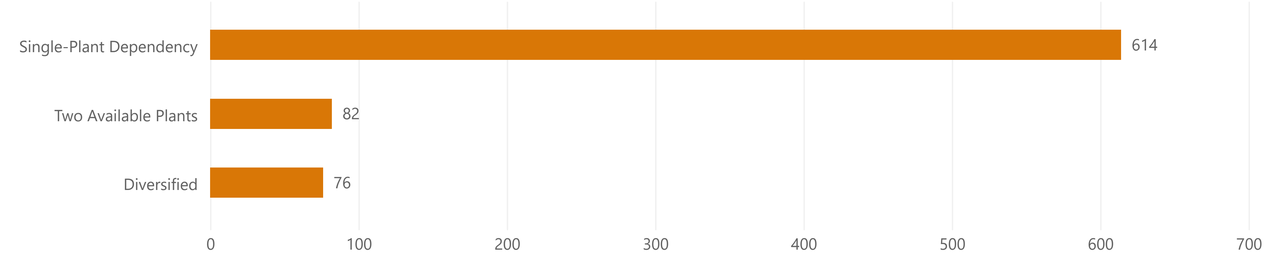
Average Orders per Customer
平均每位顧客訂單數



High-Demand Single-Plant Products
高需求單一工廠商品

product_id	total_units	total_orders	unique_customers	supply_risk
1700143	654556	10	2	Single-Plant Dependency
1700130	658352	9	2	Single-Plant Dependency
1666524	670437	19	1	Single-Plant Dependency
1683560	772283	30	1	Single-Plant Dependency
1700140	811381	17	3	Single-Plant Dependency
1667927	814076	20	1	Single-Plant Dependency
1664051	873011	37	2	Single-Plant Dependency
1700569	904493	3	2	Single-Plant Dependency
1676592	1119252	22	1	Single-Plant Dependency
1684862	3470409	16	2	Single-Plant Dependency

Product Supply Risk
商品供應風險



Key Insights

- VMI 顧客僅占全部顧客的 17.39%，但貢獻 22.67% 的訂單，平均每位顧客訂單量亦高於非 VMI 顧客，顯示 VMI 顧客具有較高的個別需求。
- 614 項商品僅由單一工廠供應，具有供應中斷風險。建議優先針對表中高需求的單一工廠商品，評估新增替代生產工廠、安全庫存或應變調度方案；同時持續追蹤 VMI 顧客需求，確保關鍵商品具備足夠產能與供應彈性。

Home
Executive Overview
Plant & Capacity
Freight & Cost
Customer & Product